

CHAITHANYA VIRUPAKSHA

Graduate Software & Machine Learning Engineer

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Right to work: eligible for the UK Graduate Route visa (from December 2026) | Open to relocation across the UK

PROFILE

MSc Advanced Computer Science graduate (Newcastle University) who builds machine-learning systems and the software around them. My dissertation showed through controlled, reproducible experiments that gradient compression changes which devices can take part in federated learning, narrowing the accuracy gap between client-selection strategies from 16.4 to 2.3 points. Hands-on with Python, PyTorch, scikit-learn, Java/Spring Boot, React, Docker, Kubernetes and Azure, with over two and a half years of industry ML experience, part-time during my degree and then 15 months full-time building and delivering models for clients. Looking for graduate roles in software, ML and cloud engineering.

TECHNICAL SKILLS

Languages: Python, Java 21, SQL, JavaScript, C++, C

Machine Learning & Data: PyTorch, scikit-learn, NumPy, Pandas, OpenCV, CNNs, federated learning, NLP (TF-IDF, LDA), experimental design, model evaluation

Software & APIs: Spring Boot 3, Quarkus, Flask, React 19, REST, OpenAPI/Swagger, JPA/Hibernate, PostgreSQL, MySQL, unit testing

Cloud & DevOps: Microsoft Azure, Azure Kubernetes Service, Docker, Kubernetes, GitHub Actions CI/CD, NGINX Ingress, Prometheus, Grafana, AWS S3, Git

Security: OAuth2/JWT, Keycloak, role-based access control, secure SDLC, threat modelling, NIST SP 800-30 risk assessment

PROJECTS

Federated Learning under Lossy Wireless Networks (MSc Dissertation)

2026

Newcastle University · Python, PyTorch, NumPy · github.com/chaithanya6699/federated-learning-packet-loss

- Built a federated-learning simulator instrumented per device and per round, modelling random packet loss, bursty packet loss and bandwidth limits across gradient-compression schemes, four client-selection policies, MNIST and CIFAR-10, and IID and non-IID data.
- Found that compression does not only cut bandwidth but decides which devices can participate: the accuracy gap between selection strategies collapsed from **16.4 to 2.3 percentage points** once updates arrived within the round deadline.
- Made every random draw a pure function of its inputs so compared configurations saw identical network conditions, removing channel noise from results; covered the framework with 50 unit tests.
- Audited my own work, found two methodological defects and documented them openly in the repository changelog rather than quietly correcting them.

Cloud-Native Bookstore Platform (team of six)

2025 – 2026

Newcastle University · React 19, Keycloak, Spring Boot 3, Azure Kubernetes Service, GitHub Actions

- Built the React front end (catalogue, book detail, cart and checkout) for a microservices platform of Catalog, Cart and Gateway services on Azure Kubernetes Service.
- Implemented browser-side authentication with Keycloak: parsed JWT claims into application state, refreshed tokens silently before expiry, and gated routes and admin actions by role.
- Made cart interactions feel instant with optimistic updates that re-fetch from the API on failure; co-authored the architecture report covering blue-green releases, autoscaling and Velero-based recovery.

AI Bias Audit Framework for Generative AI

2025 – 2026

Newcastle University · Python, scikit-learn, NLP

- Built a framework that measures gender, cultural and role-stereotype bias across several LLMs using TF-IDF, LDA topic modelling and Shannon-entropy diversity scores, with a fixed prompt set so each run is a repeatable measurement.

Tokeneer Security Risk Assessment

2025 – 2026

Newcastle University · NIST SP 800-30, threat modelling

- Mapped business risks to technical threats (biometric spoofing, privilege escalation, enrolment-station compromise) and ranked mitigations by NIST impact and likelihood, stating residual risk explicitly.

EXPERIENCE

Accurate Info Solution

Bengaluru, India

Machine Learning Engineer

May 2024 – Aug 2025

- Built and delivered machine-learning models for client projects, largely for engineering colleges, taking each from client requirements through data preparation, model training and evaluation to handover.

- Worked directly with clients to scope requirements, adapt models to their data and explain results to non-technical users.
- Progressed from intern to engineer, owning client ML projects end to end after returning to the company on graduation.

Machine Learning Intern (part-time, alongside BEng)

Sep 2022 – Dec 2023

- Built a Python and OpenCV object-detection pipeline across multiple object categories, reaching 98% classification accuracy on a held-out test set.
- Prototyped a CNN for plant-disease identification from leaf images that outperformed the rule-based baseline it replaced.

PACE Wisdom

Bengaluru, India

Full-Stack Development Intern

Sep 2023 – Mar 2024

- Rebuilt front-end components in HTML, SCSS and Bootstrap on a component-based architecture, raising Lighthouse performance scores by around 25%.
- Shipped responsive features in two-week Agile sprints alongside design and back-end engineers.

EDUCATION

MSc Advanced Computer Science — Newcastle University

2025 – 2026

Dissertation submitted September 2026; award expected December 2026

Modules: Cloud Computing, Secure Software Development, Risk & Trust Management, Generative AI for Businesses, Enterprise Software & Services, Advanced Programming in Java

BEng Computer Science — Dr. Ambedkar Institute of Technology, Bengaluru

Graduated July 2024

CGPA 8.3 / 10

CERTIFICATIONS

- Python 3 Programming — University of Michigan
- Databases and SQL for Data Science with Python

PUBLICATIONS & ACTIVITIES

- Virupaksha, C. (2026). *Privacy-Preserving Digital Twin Models for Smart Healthcare Applications*. IJDTSC, 2(1). [\[Link\]](#)
- Siddaraju, Dr., Virupaksha, C. et al. (2024). *Smart Irrigation Using Embedded AI*. IJIRCCE, 12(6).
- Bharatanatyam performer with the Indian Performing Arts Society, Newcastle University (2025 – 2026), rehearsing and performing classical Indian dance at university cultural events.
- Organised Kalarava 2024, a college festival, coordinating volunteers and logistics across multiple events.